CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents

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Abstract

Long-horizon GUI agents can retain a complete interaction trace cheaply as textual action records, but expose only a few past events to the policy in high-fidelity pixels. We formulate this as *conditional fidelity restoration*: each event persists in summary-only form and is linked to an archived screenshot, while an active visual-context budget B limits how many events may be promoted to summary-plus-image form. Recent-B spends every slot on the latest events. CAUSALCACHE instead reallocates the same B promotions over the complete trace, evicting a recent image only when a distant event has higher conditional marginal utility. Its history-gated key/value (HGKV) adapter modifies only restored history-image tokens and is exactly bypassed with no history image. Matched-budget replacement groups and per-arm-anchored difference-in-differences supervision make uniform history amplification worth zero; a budget-aware selector then chooses which summarized events to restore. On desktop, the frozen policy shows no reliable target-over-recent preference ($+0.001$ nats/token, 95% CI $[-0.024, +0.025]$), whereas HGKV creates selectivity ($+0.010$ $[+0.006, +0.014]$) inside a pre-registered drift envelope. On OSWorld-Verified, exposing historical events in high fidelity is worth $+13$–14pp over summary-only history, although same-budget allocations remain indistinguishable. Zero-shot on the 62-task construction-defined mobile memory-critical split, CAUSALCACHE improves over Recent-4 by $+8.6$pp ($[+1.6, +16.1]$), remains neutral on matched controls, and yields a split-by-method interaction of $+10.4$pp ($[+1.5, +19.6]$, p=0.022).

Introduction

GUI agents accumulate screenshots, actions, text arguments, coordinates, and interface changes over many decisions. A complete action trace persists cheaply as compressed text, but keeping every screenshot active is costly and distracting. Each past event therefore has two policy-visible fidelities: a summary-only record and, when needed, that same record with its archived screenshot reattached. The memory-control question is not only which events remain recorded, but which recorded events should be exposed again in high-fidelity pixels (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026).

Our central claim is that the object of control is *event fidelity*, not event inclusion. Existing controllers retain recent events, retrieve visually similar screenshots, or learn salience from task supervision (Zeng et al. 2026; Shi et al. 2026; Liu et al. 2026). Yet the value of promoting a summarized event back to pixels depends on which images are already active: without recent images, a distant screenshot may lack local grounding; with a saturated recent window, another image may be redundant. Moreover, the action policy must consume the reattached pixels. Our experiments show that a strong frozen policy can respond to the *presence* of history images without being reliably sensitive to their *content*, leaving an otherwise better restoration policy with nothing to improve.

CAUSALCACHE accordingly treats GUI memory as *conditional high-fidelity restoration*. Every event remains in the complete low-fidelity action trace, and an archived screenshot is reattached only when that event is promoted into the active visual context; no pixels are reconstructed from text. Under a budget of B high-fidelity history images, Recent-B promotes the latest B events. CAUSALCACHE reallocates the same B promotions over the complete trace, with recent events as the default: a distant event enters only by displacing a recent image whose conditional marginal utility it exceeds. The matched-B comparison $Q(S^*) > Q(S_{\mathrm{recent}})$ therefore isolates fidelity allocation from visual capacity. The realized number of non-recent promotions, $k = |S \setminus \text{Recent-}B|$, is a result statistic, not a preset parameter.

Learning to *use* reattached pixels must not corrupt the base agent. Full-layer adaptation simultaneously changes current-screen understanding, action syntax, and grounding. CAUSALCACHE instead trains a history-gated KV interface (HGKV) that touches only restored history-image tokens and is structurally bypassed at zero budget, with a difference-in-differences (DiD) objective in which every arm is anchored to its own frozen score: uniform amplification of any history cancels exactly, and only *selective* use of target-relevant content reduces the loss.

Our contributions are:

- a conditional fidelity-restoration formulation of GUI memory: all events persist as compressed action records, while Recent-B and CAUSALCACHE allocate the same B summary-plus-image promotions over that complete trace;
- a policy-preserving HGKV interface trained on matched-

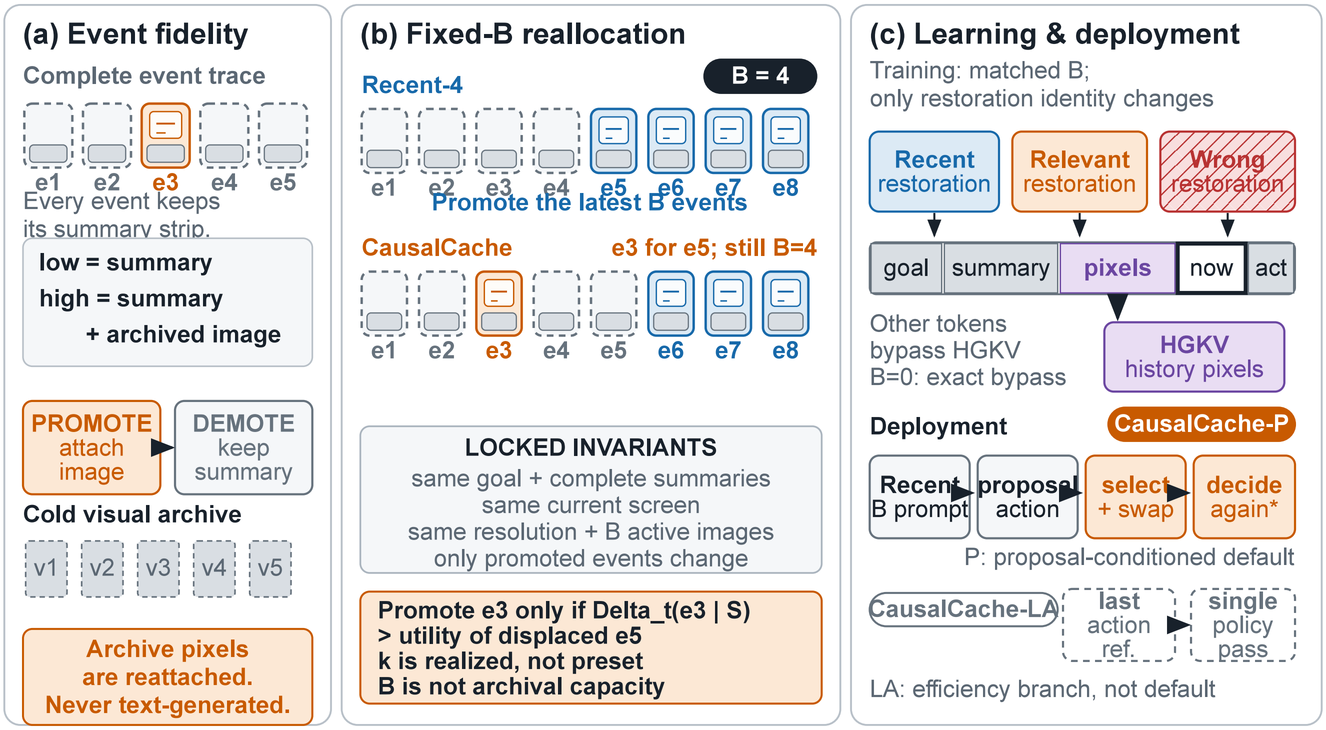

Figure 1: CausalCache treats GUI memory as conditional high-fidelity restoration. Every historical event remains in the complete action trace as a compact summary linked to an archived screenshot. Recent-B promotes the latest B events; CausalCache reallocates the same active visual slots, promoting a distant event only when its conditional marginal utility exceeds that of the recent visual realization it replaces. Matched-budget restoration arms train HGKV to modify only reattached history-image tokens, while the proposal-conditioned selector performs the same-B reallocation and re-decides only after a displacement. Pixels are retrieved from the archive, not generated from text.

budget replacement groups with per-arm-anchored DiD supervision, exact no-history parity, and pre-registered caps on recent-window and wrong-history drift, with a matched ungated control isolating the value of token gating;

- a budget-aware restoration selector over the complete summarized trace, trained on desktop data only and evaluated zero-shot on mobile and desktop benchmarks.

Related Work

Visual GUI agents and benchmarks. WebShop, WebArena, Mind2Web, WorkArena, WorkArena++, VisualWebArena, and REAL span grounded shopping, real websites, enterprise workflows, and visual web interaction (Yao et al. 2022; Zhou et al. 2024; Deng et al. 2023; Drouin et al. 2024; Boisvert et al. 2024; Koh et al. 2024; Garg et al. 2025). AITW, AndroidControl, AndroidWorld, OSWorld, and GUI-Odyssey extend executable evaluation to mobile and desktop interfaces (Rawles et al. 2023; Li et al. 2024a; Rawles et al. 2025; Xie et al. 2024; Lu et al. 2025). Screenshot-native agents learn grounding and actions directly from pixels (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). These works measure long-horizon interaction; our question is which completed event should re-enter a bounded prompt in pixels. **[Pending: Cite the MobileWorld benchmark report once its canonical reference is fixed.]**

Agent memory and reusable experience. RAG retrieves external memory, ReAct interleaves reasoning with action, and later agents preserve reflections, experience, or evolving long-term stores (Lewis et al. 2020; Yao et al. 2023; Shinn et al. 2023; Zhao et al. 2024; Zhong et al. 2024; Gutiérrez et al. 2024). ICAL retrieves multimodal programs of thought (Sarch et al. 2024). These systems write or retrieve artifacts across interactions or tasks; CausalCache instead holds a within-episode summary trace fixed and controls whether an existing event is exposed as summary only or summary plus image.

Long-horizon GUI memory. GUI-Odyssey resamples prior screenshots, MementoGUI learns memory operators, and AndroTMem retrieves causally linked anchor states (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026). Their primary variable is which item is retained or retrieved. CausalCache keeps the compressed trace fixed, controls event fidelity conditioned on the already-active visual context, and trains the interface that consumes restored pixels.

Context and visual-token compression. Prompt compressors remove textual redundancy (Jiang et al. 2023, 2024;

Pan et al. 2024); KV methods retain, evict, select, or quantize cache states (Xiao et al. 2024; Zhang et al. 2023; Liu et al. 2023; Li et al. 2024b; Liu et al. 2024; Corallo and Papotti 2024); and visual methods merge or prune tokens within an image (Bolya et al. 2023; Chen et al. 2024; Yang et al. 2025). Their unit is a token or KV position. CausalCache instead preserves the summary trace and allocates whole archived screenshots across events; HGKV controls only how promoted history-image tokens are consumed.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: a selected event regains one archived post-action image while the remaining prompt stays fixed, yielding a policy-specific behavioral surrogate rather than an environment-level causal effect. Set models parameterize coalition-conditioned scores (Zaheer et al. 2017; Lee et al. 2019), whose marginals connect to cooperative-game attribution (Shapley 1953); we use controlled restorations to train a deployable selector for one fixed GUI policy.

Method

Problem Formulation

At decision t, the agent receives instruction g, current observation x_t, and history

$$H_t = \{e_1, \dots, e_{t-1}\}, \qquad e_j = (x_j, a_j, x_{j+1}), \quad (1)$$

where each event has a compact low-fidelity representation $e_j^{\text{low}} = \tilde{e}_j$ and an available high-fidelity representation $e_j^{\text{high}} = (\tilde{e}_j, v_j)$ formed by reattaching its archived post-action screenshot v_j.

Fixed active-fidelity budget. Every prompt contains the goal g, the *complete* textual action summaries $\{\tilde{e}_1, \dots, \tilde{e}_{t-1}\}$, and the current screenshot x_t. An active visual-context budget B limits how many summarized events may be promoted by reattaching their archived pixels; it is not a limit on cold archival storage. A restoration allocation is a set $S \subseteq \{1, \dots, t-1\}$ with $|S| \leq B$. Let $Q(S)$ denote the policy's mean target-token log-likelihood of the successful next action when events in S are exposed in summary-plus-image form. The standard baseline is

$$S_{\text{recent}} = \text{Recent-}B, \quad (2)$$

the B most recent distinct events promoted to high fidelity. CausalCache targets

$$S^* = \arg \max_{\substack{S \subseteq \{1,\dots,t-1\} \\ |S| \leq B}} Q(S), \qquad Q(S^*) > Q(S_{\text{recent}}), \quad (3)$$

so recency is one candidate fidelity allocation, not a separate modality. The replacement intensity

$$k = |S \setminus \text{Recent-}B| \quad (4)$$

is a measured statistic, and the reallocation curve

$$Q_B(k) = \max_{\substack{|S| \leq B \\ |S \setminus \text{Recent-}B| = k}} Q(S), \qquad k = 0, \dots, B, \quad (5)$$

localizes where fidelity reallocation pays. All comparisons at a given B hold the image count, resolutions, summaries, and prompt structure fixed; only which summarized events are promoted differs. Cross-B comparisons are analysis, not the causal contrast. Given a partial allocation S, conditional marginals $\Delta_t(j \mid S) = Q(S \cup \{j\}) - Q(S)$ drive selection, and a replacement is justified only when the distant event's restoration marginal exceeds that of the recent visual realization it displaces. Every formally reported set utility reruns the complete restored set; we never report a sum of singleton gains as the utility of a coalition.

Policy-Preserving History-Gated KV Adapter

Let h_l be the input to layer l's attention projections. In the last eight language-model layers, CausalCache adds rank-8, α=16 low-rank residuals (Hu et al. 2022) only to key and value projections:

$$\begin{aligned} K_l &= W_l^K h_l + M_{\text{hist}}\, \Delta W_l^K h_l, \\ V_l &= W_l^V h_l + M_{\text{hist}}\, \Delta W_l^V h_l. \end{aligned} \quad (6)$$

M_{hist} is a token mask that is true only for image tokens reattached to promoted historical events. It excludes instruction tokens, event-summary text, the current observation image, and action tokens. The mask is constructed fail-closed from the multimodal token-type sequence and per-image patch geometry; any mismatch between declared images and encoded image blocks aborts the forward pass. When no historical event is promoted, the adapter context is absent and the projection hook returns the original module output, so $\pi_{\text{HG}}(\cdot \mid S{=}\varnothing) = \pi_0(\cdot \mid S{=}\varnothing)$ up to bitwise equality in our implementation.

Per-B Fidelity-Restoration Supervision

Training states are decision points of successful desktop trajectories in which the target action a_t^* recurs earlier in the same trajectory under full-action equivalence (type, arguments, and coordinates within tolerance); the screenshot preceding the earlier occurrence is the target-specific restoration candidate. For each state and each budget B we construct three matched-budget fidelity allocations with identical summaries, current screen, and target—every prompt carries exactly B history images:

- **Recent**: Recent-B, the B most recent distinct events promoted to summary-plus-image form;
- **Relevant restore**: Recent-$(B{-}1)$ + $\{v_{j+}\}$, the oldest recent visual slot replaced by the target-specific archived screenshot (age $\geq B{+}2$, strictly older than the whole window);
- **Wrong restore**: the same slot replaced by an age-matched same-trajectory screenshot whose following action is *not* equivalent to a_t^*.

Positives do not require the base action to be wrong: both information addition and evidence amplification are admissible mechanisms.

Why single-slot replacement (k=1). Training teaches only the k=1 replacement interface, and this is an empirically grounded interface choice rather than a structural limit. On the demand side, multi-frame needs are rare even where archived pixels matter most: under the strictest *image-only* criterion—the required value appears in neither the goal, the current frame, the recent window, nor any completed step's action text—mining 1,326 GUI-Odyssey trajectories yields 96 image-only decision points at B=4, of which only 2 require two distinct evidence frames simultaneously (3 of 239 at B=1); on the desktop corpus, 75% of eligible states admit exactly one recurrence witness. On the utility side, a matched-budget probe under the official multi-turn protocol shows the first replacement is worth its slot (+0.013, 95% CI $[+0.005, +0.021]$ frozen log-likelihood margin) while paired increments of a second and third replacement are indistinguishable from zero or negative (+0.001 and -0.011, n.s.)—under a fixed budget each further distant promotion must evict the next-most-recent visual realization, and its measured value no longer covers that price. Larger-k allocations nonetheless remain expressible at deployment: the selector composes exactly-B allocations with exact re-scoring, defaulting unclaimed slots to recent events, so k stays a result statistic. Because supervision is annotation-free—labels are teacher-forced counterfactual scores of the current policy on its own trajectories—a domain that does exhibit multi-frame demand (detectable by the same probe) requires only regenerating the corpus with $k \geq 2$ restoration arms and re-running the identical trainer: a data-mix change, not a method change.

Let ℓ denote mean target log-likelihood, superscripted by adapter state (active or frozen bypass). Each arm is anchored to *its own* frozen score:

$$A_s = \ell^{\text{on}}_{\text{relevant}} - \ell^{\text{off}}_{\text{relevant}}, \qquad A_r = \ell^{\text{on}}_{\text{recent}} - \ell^{\text{off}}_{\text{recent}}, \\ A_n = \ell^{\text{on}}_{\text{wrong}} - \ell^{\text{off}}_{\text{wrong}}, \tag{7}$$

and the DiD objective is

$$\begin{aligned}\mathcal{L} = {} & [m - (A_s - A_r)]_+ + [m - A_s]_+ \\ & + [m - (A_s - A_n)]_+ \\ & + \lambda_{\text{cap}}\big([|A_r| - \epsilon]_+ + [|A_n| - \epsilon]_+\big) + \lambda\|\Delta W\|_2^2,\end{aligned} \tag{8}$$

with m=0.01, ϵ=0.02, λ_{cap}=2, λ=10^{-4}, and no action cross-entropy. On the identity adapter every increment is zero, so $A_s - A_r$ is exactly zero: the frozen policy's own preference for target-relevant restorations cannot masquerade as adapter skill, and uniformly amplifying all restored pixels buys nothing. The dead-zone caps bound recent-window and wrong-history drift with gradients on the same scale as the selection hinges. Losses and evaluation are stratified by B; a pooled mean could hide a method that helps at B=4 but fails at B=1.

Success-Anchored Utility and Selector Supervision

After the history interface is frozen, selector labels are true policy utilities: for each state, singleton utilities for the *entire* summarized event pool (recent events included)—an exhaustive B=1 oracle—the matched $Q(\text{Recent-}B)$ anchors, and, for larger budgets, set utilities along beam-3 teacher paths over a singleton-shortlisted pool. Every set is fully re-scored as a complete prompt through the frozen HGKV policy; set utility is never approximated by summing singleton utilities.

Budget-Aware Restoration Selector

The selector is a single set-conditional marginal scorer $\widehat{\Delta}_t(j \mid S)$ trained on the true marginals implied by the label table (singleton edges give S=$\varnothing$; each beam edge gives its parent set), with a joint regression and within-state ranking loss. Its input encodes, explicitly, candidate age and recency-window membership, target-recurrence witness signals with coordinate distance, action family, frame-deduplication multiplicity, trajectory shape, and set context (selected-set size, overlap and age gaps between the candidate and the already-selected events).

Deployment composes exactly-B fidelity allocations by beam search initialized at $\text{Recent-}B$: every summarized event is a candidate, and a distant event is promoted only by displacing the recent visual realization whose predicted marginal it beats, realizing Eq. 3 as a relative comparison with no learned stopping threshold—declining every replacement recovers Recent-B exactly. Selector labels cover $B \in \{1, 2, 4\}$; the realized replacement intensity k (Eq. 4) is reported per budget, never preset.

Reference actions and the two deployment modes. Four of the selector's input features are *witness* features: they test whether the action taken right after some archived event matches a reference action. At training time the reference is the gold teacher-forced target—an offline oracle that upper-bounds what any online reference can provide but does not exist at deployment. We therefore instantiate two online modes that share every trained weight and differ only in the reference. **CausalCache-P** (default) uses the current policy's tentative action: the policy first acts on the Recent-B prompt, the tentative action serves as a platform-calibrated reference, and the policy re-decides on the re-allocated prompt when a recent visual realization is displaced. **CausalCache-LA** replaces the proposal with *the action just executed* at step $t-1$, eliminating the second policy call: memory is selected before the single generation, adding only the selector's own cost ($<$ 10 ms). On held-out desktop decision groups the two modes are statistically indistinguishable (LA +0.015 $[+0.006, +0.025]$, gold-reference +0.011 $[+0.001, +0.021]$ over Recent-4), but LA fails to transfer to the unseen mobile benchmark, where P attains the higher mean although the current P-versus-LA interval crosses zero. We therefore treat LA as an efficiency variant rather than the primary method; the directional result is consistent with the reference action being one bottleneck for cross-platform transfer.

A ladder of weaker references isolates what the action signal buys. *No reference* (witness features zeroed) keeps B=2 significant (+0.016 $[+0.003, +0.030]$) but loses B=4 (+0.009 $[-0.002, +0.020]$) while more than tripling full-replacement allocations (k=4: 11% → 34%)—without any action signal the selector cannot tell which summarized event merits visual restoration and over-replaces. *Instruction-only intent* (eight frozen-embedding cosine features between the instruction, each candidate's action line, and recent behavior) repairs the over-replacement pathology (k=4 back to 12%) yet stays non-significant at B=4 ($+0.005$ $[-0.007, +0.016]$). Action-conditioned references thus dominate instruction-only and reference-free variants; the relative order of the action-conditioned references themselves depends on platform and protocol, which is precisely why the deployed default re-grounds the reference in the current policy's own proposal.

Experiments

Experimental Setup

Training Data and Model Selection. Policy adaptation and selector training use only desktop data: successful AgentNet/OpenCUA Ubuntu trajectories (5,000 screened, 2,293 successful, 6,003 decision points) yield fixed-budget replacement groups at B=1/2/4 of 969/764/470 (2,203 in total; 1,774 training units and 211 development groups, trajectory-disjoint; click 52–56%, plus key/type/drag/scroll), complemented by a high-precision OSWorld witness seed mined from official successful trajectories. A further 160 groups at B=8 are constructed but deliberately excluded from training, so B=8 evaluates out-of-training budget extrapolation. Checkpoints and thresholds are selected only on the desktop development split under pre-registered gates: positive DiD selection with a positive bootstrap lower bound, and drift caps $|A_r|, |A_n| < \epsilon$. No mobile benchmark data touches training or selection.

The Fixed-Budget Fidelity-Restoration Design. Figure 1 defines the core comparison and the two learned components. Every compared allocation carries exactly B history images with identical resolutions, summaries, and prompt structure, so the contrast isolates *which summarized events are promoted to high fidelity*. B=4 is the primary deployment-like setting; the realized number of non-recent promotions k is measured rather than preset.

Policies and Fidelity-Allocation Baselines. The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). We compare four policy rows under identical data, steps, and cadence:

- **Frozen**: no trainable adapter;
- **Ungated KV**: the same last-eight layers, K/V targets, rank, and alpha as HGKV but active on all tokens;
- **HGKV (ours)**: token-gated, structurally bypassed at B=0.

Fidelity-allocation baselines are Recent-B, Random, OCR/RGB similarity, and the learned budget-aware restoration selector (exact-B, recent fallback); a beam oracle over the candidate pool bounds headroom. Every method receives the same summaries and the same active image budget.

Zero-Shot Benchmarks and Metrics. The policy interface and selector, trained on desktop data only, are evaluated (i) on held-out desktop decision groups (offline, per-B), (ii) zero-shot on an uncontaminated OSWorld task roster (Xie et al. 2024) with official evaluator scores (tasks whose trajectories seeded the witness corpus are excluded from zero-shot claims and reported separately as in-domain diagnostics), and (iii) zero-shot on a cross-app mobile memory benchmark of 117 runnable tasks (62 cross-app memory candidates, 55 single-app controls), which contributes no data to any training or selection decision. **[Pending: Fix the MobileWorld citation and the final roster hashes.]** Offline metrics: mean target log-likelihood margins, full-action equivalence (type, arguments, coordinates at τ=25 of a $[0, 999]$ grid; $\tau \in \{5, 10\}$ as sensitivity), repair and amplification rates, wrong-history harm, parser validity, and active history-image count. Closed-loop metrics: paired task success with cluster bootstrap, step counts, and wall time.

Results

Selectivity Is Not Inherited from the Frozen Policy. On the desktop development split at B=1 (94 groups, 77 trajectories), the choice of recent baseline matters. Against a *redundant* baseline that promotes a byte-identical copy of the current screen (the deployment recent-selector's behavior), the frozen policy prefers the target-relevant archived screenshot by $+0.0333$ nats/token (95% CI $[+0.0118, +0.0567]$). Against the *informative* true previous frame, however, its preference vanishes: $+0.0013$ $[-0.0242, +0.0253]$, not significant (94/94 probes have a previous frame that differs from the current screen). To the frozen policy, promoting a task-relevant distant event and retaining the recent event it would displace are worth about the same; profitable fidelity reallocation must be learned. Under the official multi-turn protocol the frozen replacement effect is $+0.079/+0.026/-0.000/-0.031$ at B=1/2/4/8: whatever native preference exists decays with the window and reverses, while the learned interface keeps a positive selection effect across the whole budget axis.

Selective Use Within a Drift Envelope. Table 3 reports the pre-registered desktop DiD gate at B=1. All three adapters are trained for 150 matched steps; checkpoints are selected by the frozen rule (all gates passed, maximal DiD selection, earliest tie-break). HGKV passes every gate from step 50 onward: its selected checkpoint improves DiD selection to $+0.0111$ $[+0.0074, +0.0150]$ against the redundant-duplicate recent control, and a re-scoring probe confirms $+0.0095$ $[+0.0056, +0.0138]$ (60/94 groups) against the informative true-previous-frame control, while holding recent-only and wrong-history drift well inside ϵ=0.02; its no-history path is bitwise identical to the frozen policy under randomized nonzero adapter weights. The layer- and rank-matched ungated control also learns selectivity ($+0.0076$), confirming that the last-eight K/V position carries signal, but token gating adds a paired per-group advantage of $+0.0035$ (95% CI $[+0.0007, +0.0062]$, 56/94 groups) while holding drift further from the caps. Gating is what buys selectivity *inside* the envelope.

Fidelity Reallocation and Zero-Shot Transfer. Per-B DiD gates all pass for HGKV at every budget, including the untrained B=8 extension: $\mathrm{did}_{\mathrm{select}} = +0.0224$ $[+0.0162, +0.0294]$, $+0.0164$ $[+0.0110, +0.0221]$, $+0.0109$ $[+0.0070, +0.0154]$, and

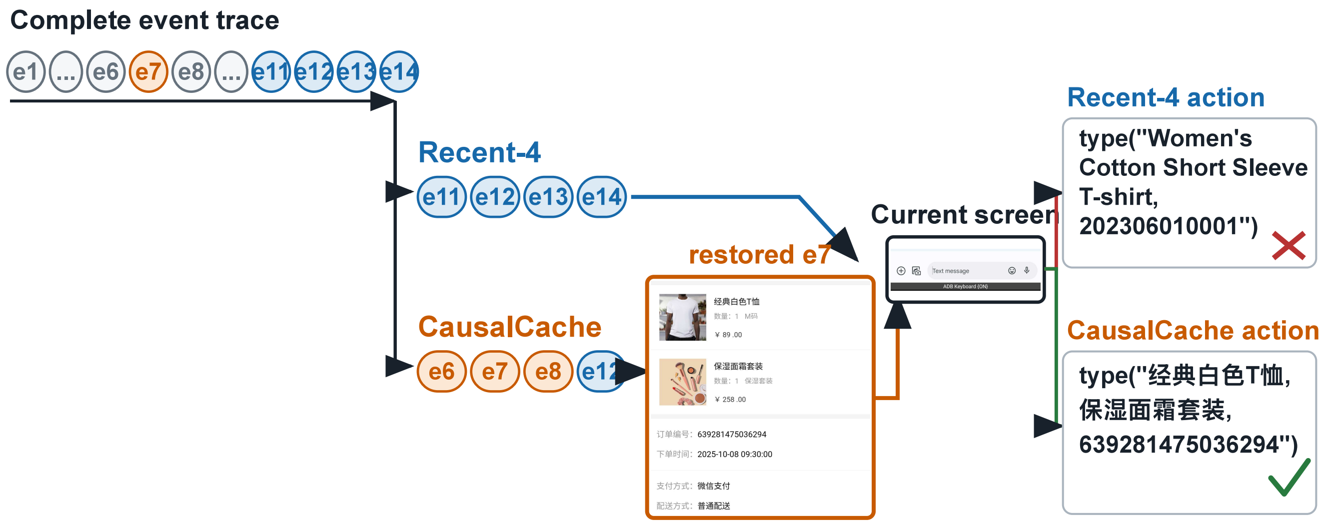

Figure 2: A real cross-app `CartInfoNotificationTask` episode. The agent must inspect items awaiting shipment in TaoDian, then send an SMS containing only the product names and order number. Both rollouts retain the complete summary trace and use four active history-image slots. Recent-4 exposes $[e11, e12, e13, e14]$. CausalCache-P instead exposes $[e6, e7, e8, e12]$, restoring the enlarged $e7$ order screenshot before the shared message-composition stage. Recent-4 types the wrong item and order number and fails; CausalCache types the visually recovered values and succeeds. This single episode illustrates the mechanism rather than an aggregate effect.

Fixed-budget fidelity allocation (offline desktop heldout; primary setting $B = 4$)

Policy	Allocation	B=1	B=2	B=4	B=8†	Realized k	Notes
Frozen	Recent-B	0	0	0	0	0	reference
Frozen	oldest slot → archived	+.079	+.026	−.000	−.031	1	
HGKV	Recent-B	+.002	+.005	+.007	+.009	0	drift-capped
HGKV	oldest slot → archived	+.103	+.047	+.018	−.016	1	gate: Tab. 3
HGKV	selector, LA reference (deployed single-pass)	+.039	+.025	+.015	−.005	measured	71% displace ≥1 at B=4
HGKV	selector, gold reference (offline upper bd.)	+.044	+.025	+.011	−.002	measured	teacher-forced oracle

Table 1: Fixed-budget fidelity-allocation results. Each cell is a matched-count contrast on the desktop heldout split; every allocation promotes exactly B events to summary-plus-image form, and the learned selector may keep every promotion on the recent window (k=0); $^\dagger B$=8 is excluded from training and reports budget extrapolation. Cross-B comparisons are analysis only.

Method	OSWorld-Verified (361)	MobileWorld (117)	Extra policy calls
Frozen, summary-only (B=0)	19.9	28.2 (3 runs)	0
Frozen + Recent-4	32.7	29.9	0
HGKV + Recent-4	33.0	30.2 (3 runs)	0
CausalCache-LA (single-pass)	**34.1**	29.1	0
CausalCache-P (default)	32.7	**33.9** (3 runs)	conditional

Table 2: Closed-loop success rates (%). OSWorld-Verified: official no-GDrive roster and official 15-step budget, one run per arm. MobileWorld: zero-shot three-run means. High-fidelity arms beat summary-only memory on desktop but remain pairwise indistinguishable; CausalCache-P is best on mobile. Full paired statistics and per-round results are in the supplementary document.

$+0.0063$ $[+0.0018, +0.0115]$ for B=1/2/4/8, with recent-window and wrong-history drift within the pre-registered caps throughout. A layer- and rank-matched ungated KV control also passes its gates but attains uniformly smaller selection effects ($+0.0177/+0.0139/+0.0075/+0.0047$) and drifts toward the cap at B=8 ($|A_r|$=0.016 vs. 0.009 for HGKV): token gating adds selective gain and containment beyond matched capacity. The frozen replacement effect

Adapter (selected step)	DiD select	$\lvert A_r \rvert$	Wrong drift	Gates
HGKV (s150)	$+$**0.0111**	0.0087	0.0085	pass
Ungated KV (s125)	$+0.0076$	0.0090	0.0063	pass

Table 3: Desktop DiD gate on 94 development groups (B=1; 2,000 episode-clustered bootstrap replicates). DiD select = $(A_s - A_r)$; caps require $|A_r|, |A_n| < 0.02$. HGKV's DiD CI is $[+0.0074, +0.0150]$; HGKV B0 parity is bitwise exact.

decays from $+0.079$ at B=1 to -0.031 at B=8, while the learned end-to-end selector beats Recent-B at matched image counts by $+0.030$ $[+0.019, +0.041]$ (B=2) and $+0.017$ $[+0.007, +0.028]$ (B=4), keeping the full recent window in 19% of B=4 states.

Mobile closed loop (zero-shot). The 117-task mobile benchmark — which contributes no training or selection signal — is our primary test of *allocation*, because its roster is constructed with a memory-critical split. The full-roster aggregates are secondary: CausalCache-P attains the highest overall success rate, 33.9% (three runs: 39/42/38 of 117) against 30.2% for HGKV+Recent-4 (three runs: 34/37/35; paired +3.7pp, 95% CI $[-0.9, +8.6]$, per-round $+4.3/+4.3/+2.6$pp) and 28.2% for the frozen summary-only baseline (three runs: 33/32/34; +5.7pp, $[-0.6, +12.0]$). The decomposition is clean: frozen Recent-4 (29.9%), HGKV+Recent-4 (30.3%), and the summary-only baseline are statistically indistinguishable — the behavioral counterpart of the drift cap — so the gain is carried by *which summarized events are promoted*, not by the adapter or by recency alone. A full recent-dose sweep of the frozen policy makes the last point explicit: across B=0/1/2/3/4 the success rate is 28.2/26.5/26.5/29.1/29.9%, non-monotone with every paired contrast against B=0 crossing zero — on this benchmark, promoting additional recent events alone buys nothing, so fidelity *allocation* rather than active visual budget *size* is the operative variable. The primary closed-loop result is the memory-critical split, defined at benchmark construction: 62 cross-application tasks whose target information surfaces in an application visited *earlier* in the episode — after the app switch, the Recent-B window no longer contains that evidence by construction — and 55 matched single-application controls with no such dependency. On the memory-critical stratum the same-budget contrast is decisive: CausalCache-P 28.0% vs. Recent-4 19.4%, +8.6pp (95% CI $[+1.6, +16.1]$); on the controls the method is correctly inert (40.6% vs. 42.4%, -1.8pp $[-6.7, +3.6]$); the split-by-method interaction is +10.4pp (95% CI $[+1.5, +19.6]$, two-sided p=0.022). The value of reallocation is conditional on distant visual dependency: the modest full-roster mean is composition, not absence of effect, since 47% of the roster is control tasks on which no high-fidelity restoration method should help.

As a complementary outcome-defined analysis, we call tasks that the frozen summary-only policy cannot complete within the official budget *memory-demanding*, and report $\Delta_{\text{long}} = P(\text{success} \mid \text{CausalCache}, \text{long}) - P(\text{success} \mid \text{baseline}, \text{long})$ on that stratum. Across 95 such mobile tasks, CausalCache-P gains 7.0pp over summary-only memory (95% CI $[+0.4, +14.0]$), while adding nothing on the 22 tasks it already solves. The CausalCache-LA efficiency variant does not transfer (-4.8pp vs. P, $[-10.5, +0.6]$), localizing the cross-platform signal to the proposal reference. These post-hoc strata are secondary to the construction-defined memory-critical split.

Desktop closed loop. On OSWorld-Verified (361 tasks, official no-GDrive roster, docker provider, and 15-step budget), all high-fidelity arms reach 32.7–34.1%, each +13–14pp over summary-only memory (19.9%, paired 95% CIs excluding zero), while same-budget allocations remain indistinguishable. OSWorld therefore supports high-fidelity *availability*; conditional *allocation* is tested by the mobile memory-critical split. HGKV is neutral on matched Recent-4 allocations (+0.3pp, p=0.96). The P arm adds a median eight policy calls and 20.6 model-seconds, or 7% of median end-to-end task time.

Ablations and Analyses

Policy-side per-B gates and the frozen recent-dose sweep appear in Results. At matched B=2/4, the selector beats Recent-B by $+0.025/+0.015$; random exact-B and RGB-similarity restoration cross zero, while inverting the learned score is significantly harmful ($-0.016/-0.009$). Removing recency-membership features returns performance to the noise band. Thus the gain comes from the learned conditional ranking, not sparse exposure or visual similarity alone; complete CIs and the witness-reference ladder are reported in the supplementary document.

Limitations and Conclusion

We control an active context budget, not archive capacity: both policies expose exactly B summary-plus-image events, and CausalCache reattaches rather than generates pixels. We claim only that some summarized events have greater conditional marginal utility than the recent realization they displace. "Causal" denotes controlled prompt interventions, not structural causality, and teacher-forced margins must be judged by closed-loop success. We report a true-previous-frame control for the duplicated training-time B=1 slot; pretraining contamination remains possible.

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